Database Schema

Tables

The database has 9 tables and 10 foreign keys. The schema is **not** created automatically by the application; it must exist in any new database.

cohort

Named groups of users (for example "Cohort A - Brisbane North").

Column	Type	Null	Key / extra
name	varchar(128)	no	primary key

conversation

A chat: type is `cohort`, `group` or `direct`.

Column	Type	Null	Key / extra
conversation_id	int	no	primary key, auto_increment
type	enum('cohort','group','direct')	no	-
name	varchar(255)	yes	-
created_by	int	no	indexed
created_at	datetime	no	DEFAULT_GENERATED

conversation_member

Which users belong to a conversation.

Column	Type	Null	Key / extra
conversation_member_id	int	no	primary key, auto_increment
conversation_id	int	no	indexed
user_id	int	no	indexed
joined_at	datetime	no	DEFAULT_GENERATED

event

Calendar events created by administrators.

Column	Type	Null	Key / extra
event_id	int	no	primary key, auto_increment
user_id	int	no	indexed
name	varchar(45)	no	-
datetime	datetime	no	-
location	varchar(200)	no	-
description	varchar(500)	no	-

event_member

Which users are assigned to an event and whether they confirmed.

Column	Type	Null	Key / extra
event_id	int	no	primary key
user_id	int	no	primary key
confirmed	tinyint	no	-

folder

Resource folders; `parent_id` NULL marks the root. `cohorts` and `access_level` control visibility.

Column	Type	Null	Key / extra
folder_id	int	no	primary key, auto_increment
name	varchar(128)	no	-
parent_id	int	yes	indexed
cohorts	json	no	DEFAULT_GENERATED
access_level	enum('student','ranger','admin')	no	-
created	datetime	no	-

message

Individual chat messages.

Column	Type	Null	Key / extra
message_id	int	no	primary key, auto_increment
conversation_id	int	no	indexed
sender_id	int	no	indexed
text	text	no	-
created_at	datetime	no	DEFAULT_GENERATED

resource

Uploaded files. The file itself is in S3 under `resource-hub/<resource_id>`.

Column	Type	Null	Key / extra
resource_id	int	no	primary key, auto_increment
type	enum('video','audio','image','pdf','document','spreadsheet','presentation')	no	-
title	varchar(255)	no	-
uploader_id	int	no	indexed
extension	varchar(256)	no	-
mime_type	varchar(128)	yes	-
parent_id	int	no	indexed

Column	Type	Null	Key / extra
cohorts	json	no	DEFAULT_GENERATED
access_level	enum('student','ranger','admin')	no	-
status	enum('pending','success','failed')	no	-
created	datetime	no	-

user

Accounts. `status` is Pending until an administrator approves; `role` is admin, ranger or student.

Column	Type	Null	Key / extra
user_id	int	no	primary key, auto_increment
name	varchar(64)	no	-
email	varchar(64)	yes	unique
mobile	varchar(16)	yes	unique
password_hash	varchar(256)	no	-
status	varchar(16)	no	-
created	datetime	no	-
role	enum('admin','staff','ranger','student')	no	-
cohort	varchar(64)	yes	-

Relationships

Table	Column	References table	References column
conversation	created_by	user	user_id
conversation_member	conversation_id	conversation	conversation_id
conversation_member	user_id	user	user_id
event	user_id	user	user_id
event_member	event_id	event	event_id
event_member	user_id	user	user_id
message	conversation_id	conversation	conversation_id
message	sender_id	user	user_id
resource	parent_id	folder	folder_id
resource	uploader_id	user	user_id